1 Introduction

At the heart of the paper's quantification, the estimated efficiency gaps enter directly as a wedge on public spending: an efficiency gap of, say, 0.35 implies that 35 percent of each unit spent does not reach the productive public capital stock—lost to over-invoicing, poor project selection, weak implementation, or inadequate maintenance. The appeal of this formulation is that it separates the level of spending from its productivity—a country spending as much as its peers but with a larger gap wastes a higher share—so that closing the gap raises output without additional budget resources.

2 Public Spending in SACU: Stylized Facts

This section documents three features of SACU public spending that together motivate the focus on efficiency and composition: spending is high but tilted toward consumption rather than investment; it is rigid and procyclical with SACU revenue; and outcomes lag spending inputs, pointing to a technical-efficiency wedge.

2.1 Spending is High and Tilted Toward Consumption

Across the five SACU members, general government expenditure averages between 31 and 52 percent of GDP (Figure 1)—a level comparable to advanced economies and substantially above most peer emerging markets. The composition, however, is heavily tilted toward current spending. The public-sector wage bill is the dominant single line in every SACU budget: it averaged 33 percent of total expenditure in Lesotho during 2017–22, 34 percent in South Africa, 38 percent in Eswatini, 42 percent in Namibia, and 40 percent in Botswana (Figure 2). These shares are well above the 24 percent average in advanced economies and the 28 percent average in sub-Saharan Africa.

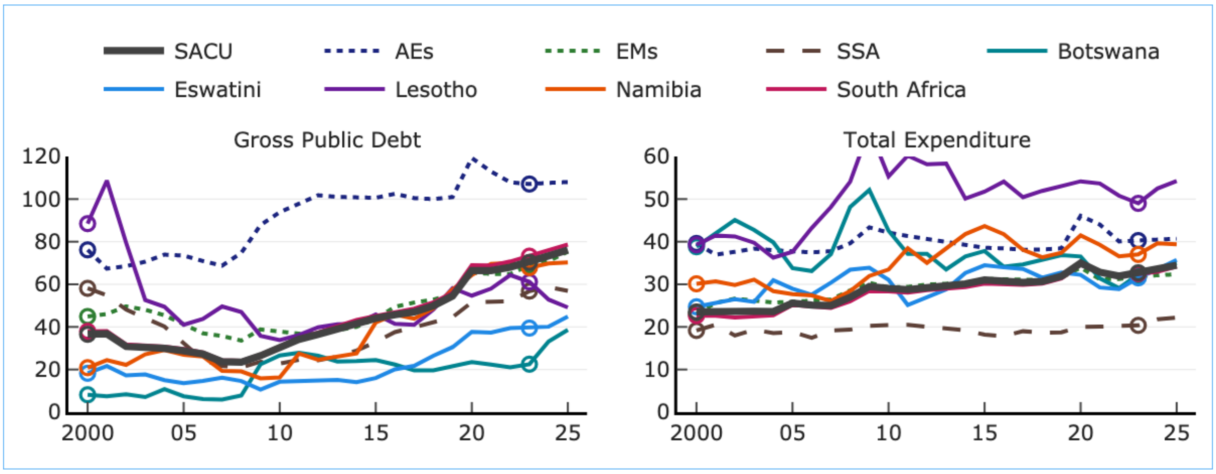
The level of the wage bill matters because it displaces discretionary spending on growth-enhancing categories. In Lesotho, the wage bill averaged 17 percent of GDP over the last fifteen years, reaching 18 percent in 2018–23—far above the 11.7 percent average for SACU countries and the 8 percent average for lower-middle-income countries (World Bank 2025b). In Eswatini, the wage bill reached 10.4 percent of GDP in 2023, against 7.0 percent for regional comparators and 6.9 percent for aspirational peers (World Bank 2025a). South Africa's public wage bill has risen by 2.5 percentage points of GDP over two decades and now represents about a third of public spending and 40 percent of total revenue (IMF 2025e).

A second feature compounds the level effect: the public-private wage premium. The right panel of Figure 2 plots, for each country in the IMF Government-Private Sector Compensation Premium Dataset, the percentage by which public-sector compensation exceeds private-sector compensation for workers of comparable qualifications. In SACU, the premium ranges between about 30 percent (Lesotho) and around 50 percent (South Africa, Eswatini, Botswana). These are among the highest premia observed globally. The combination of a large public payroll and a high premium implies that the fiscal envelope devoted to wages does less work per currency unit than it would in a better-calibrated system.

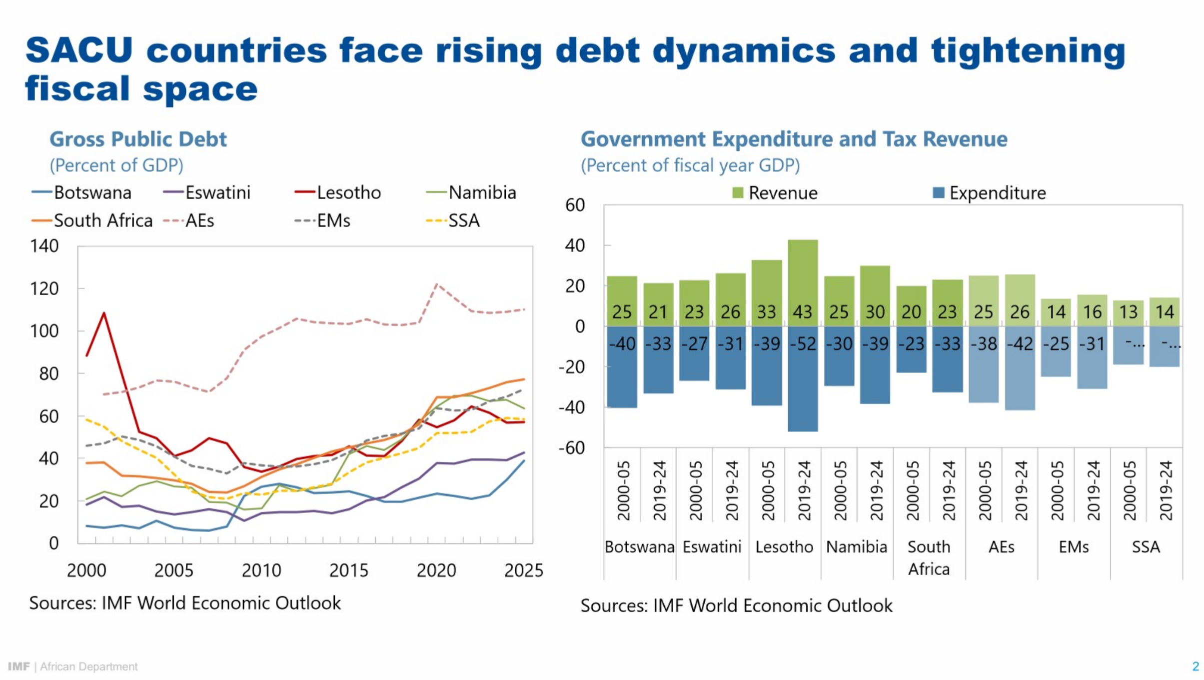
2.2 Spending is Rigid and Procyclical with SACU Revenue

The second stylized fact is that SACU spending is rigid—that is, spending in period t is highly autocorrelated with spending in period $t-1$. The Fiscal Monitor measures rigidity as the one-year positive autocorrelation of spending components over five-year rolling windows, with a score between 0 (fully flexible) and 1 (fully rigid). Figure 3 reports the distribution of this index for SACU countries and for three peer groups—sub-Saharan Africa (excluding SACU), emerging markets, and low-income developing countries—over two sub-periods.

Figure 1: Public-Sector Debt, Expenditure and Revenue Dynamics in SACU
(New Version (Plotly replication))

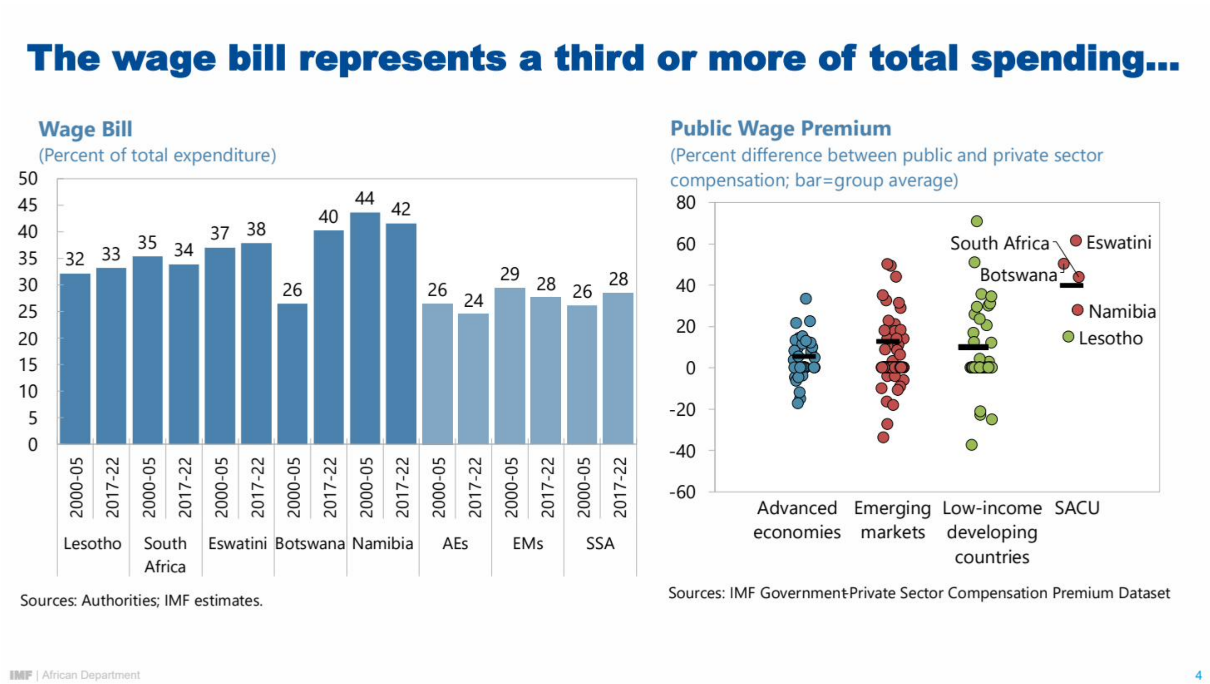


(Old Version (Original document snapshot))



Source: Reproduced from IMF (2026).
Note: General government expenditure and debt profiles for SACU countries.

Figure 2: Public-Sector Wage Bill and Public Wage Premium in SACU
(SACU wage bills are well above peer groups, and public wage premia are among the largest observed)



Source: Reproduced from IMF (2026).
Note: *Left panel*: wage bill share of total expenditure, national authorities and IMF staff estimates. *Right panel*: IMF Government-Private Sector Compensation Premium Dataset.

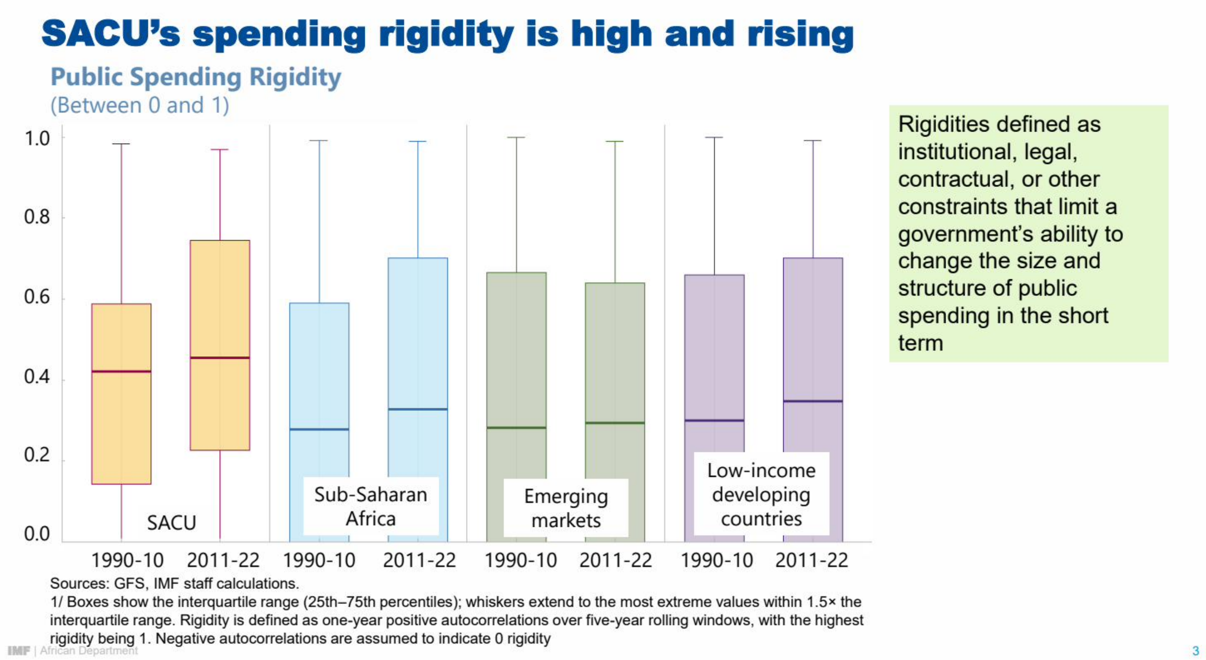
Three patterns are visible. First, SACU's median rigidity rose from about 0.42 in 1990–2010 to about 0.45 in 2011–2022, while peer-group medians were between 0.28 and 0.35. Second, SACU's interquartile range is well above all three peer groups in both periods, confirming that rigidity is not driven by a single outlier. Third, the rigidity of SACU spending rose while the peer groups' rigidity was relatively stable, suggesting that the structural shift has a SACU-specific driver.

That driver is the asymmetric response of spending to SACU revenue. When SACU transfers rise, the wage bill, transfers to state-owned enterprises, and current transfers tend to rise with them. When SACU transfers fall, the rigid components do not adjust downward at the same speed. The Lesotho Economic Update (World Bank 2025b) documents this pattern directly, noting "that the public sector wage bill increases when SACU revenues rise but does not fall when they fall". The result is a ratchet: each SACU revenue boom raises the permanent spending level, tightening rigidity and narrowing future adjustment options.

2.3 Outcomes Lag Spending Inputs

The third stylized fact is that outcomes lag spending inputs. This motivates the technical-efficiency analysis in Section 3. Namibia's Article IV (IMF 2025d) documents persistent gaps between high levels of education and health spending and the associated outcome indicators (learning assessments, life expectancy, infant mortality). Botswana's Article IV surveillance reports (IMF 2025c) identifies execution and outcome gaps despite an adequate resource envelope, attributing them in part to weaknesses in project appraisal and implementation. Lesotho's Economic Update (World Bank 2025b) describes how weaknesses in public financial management, public investment, and public procurement blunt the impact of each loti spent. Eswatini's Public Finance Review (World Bank 2025a) finds that despite wage spending dominating the social-sector budget, education and health outcomes lag aspirational peers. South Africa's Article

Figure 3: Public Spending Rigidity, SACU vs Peer Groups
(Spending rigidity in SACU is the highest among peer groups and has risen over time)



Source: Reproduced from IMF (2026), based on IMF Government Finance Statistics and staff calculations.
Note: Boxes show the interquartile range (25th–75th percentiles); whiskers extend to the most extreme values within 1.5× the interquartile range. Rigidity is defined as the one-year positive autocorrelation of spending components over five-year rolling windows, scaled 0 to 1; negative autocorrelations are set to 0.

IV (IMF 2025e) notes that procurement-system challenges, SOE transfers, and infrastructure-delivery gaps compound the efficiency shortfall.

The consistency of this pattern across the five countries—high spending, composition tilted toward consumption, rising rigidity, outcomes that lag inputs—points strongly to a technical-efficiency wedge as a binding constraint on the growth return from public spending. The next section measures that wedge.

3 Measuring Spending Efficiency: Methodology and SACU Results

This section explains the stochastic-frontier methodology used in the Fiscal Monitor (IMF 2025a) and reports the SACU-specific efficiency-gap estimates that enter the paper's simulations. Because these gaps conflate several underlying factors—including institutional capacity, governance quality, and country-specific shocks—they do not, on their own, identify the mechanisms by which resources are lost. To address this, Section 4 links the measured gaps to institutional evidence from the Public Investment Management Assessments.

3.1 Methodology

This paper relies on the spending-efficiency measures established in Bańkowski, Cho, and Sher (2026), which underpin the October 2025 Fiscal Monitor (IMF 2025a). Their approach compares public spending (the input) with the socioeconomic outcomes it is meant to deliver (the output) in each of the four spending sectors covered by the framework—public investment, education, health, and research and development.[1] Stochastic frontier analysis (SFA) estimates a best-practice production function from a cross-country panel and measures each country's efficiency as its distance below that frontier. Unlike non-parametric frontiers such as data envelopment analysis, SFA separates genuine inefficiency from statistical noise. It also handles persistent structural differences between countries—such as being landlocked or having a large private health sector. These are absorbed separately, through the four-component ("true random effects") estimator of Filippini and Greene (2016). The estimator distinguishes time-invariant structural heterogeneity from persistent inefficiency, so the former is not mislabelled as the latter.

Inputs are measured as a five-year moving average of real public spending per capita in purchasing-power-parity terms; outputs combine multiple quantity and quality indicators for each sector (for public investment, for example, road and electricity access alongside infrastructure-quality ratings; for education, enrolment, completion, and learning measures; for health, life expectancy, immunization, and health-system capacity). To avoid sensitivity to any single indicator, scores are averaged across all combinations of the available indicators. Because the estimator is time-varying, it delivers a separate score for each individual country in each individual year, rather than a single cross-sectional snapshot. Each score lies between 0 and 1, with 1 denoting a country on the frontier. Throughout this paper we work with the complementary *efficiency gap*—one minus the score—so that 0 denotes full efficiency and higher values denote larger shortfalls. Full methodological detail is given in Bańkowski, Cho, and Sher (2026) and the Fiscal Monitor Online Annex (IMF 2025b).

3.2 SACU Results

Efficiency gaps persist across sectors in SACU. Figure 4 reports the Fiscal Monitor efficiency gaps for the five SACU countries in three sectors—infrastructure, health, and education—at two points in time (2000 and 2023), alongside the unweighted averages for advanced economies, emerging markets, and sub-Saharan Africa. Three patterns stand out. In every sector, SACU gaps sit above the advanced-economy average and close to the emerging-market level. Infrastructure has improved the most since 2000. Health gaps remain the largest and most persistent, while education has stayed broadly flat even when it deteriorated across advanced economies and emerging markets.

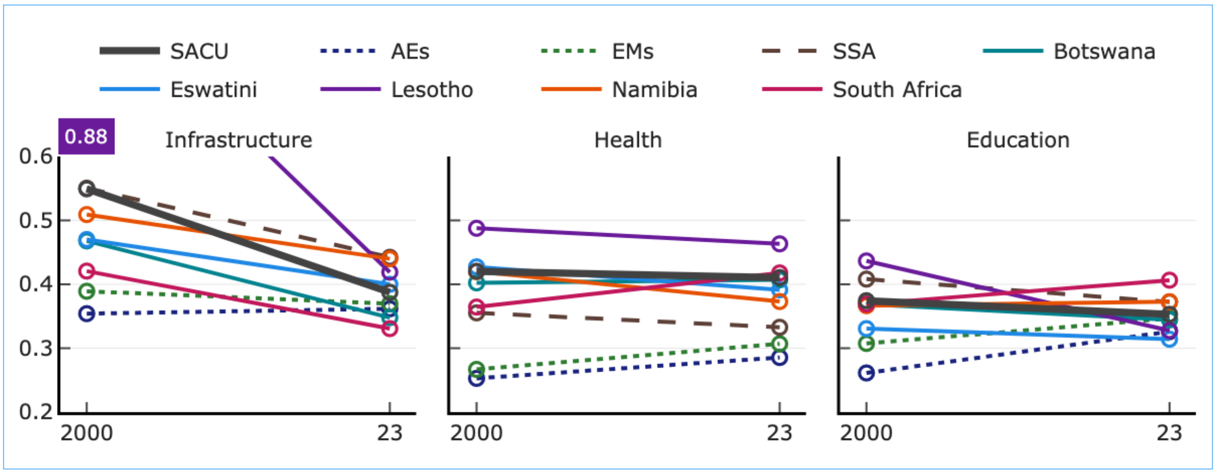
Infrastructure gaps have narrowed but remain wider than peers. The left panel of Figure 4 shows that all five SACU countries closed their infrastructure efficiency gaps substantially between 2000 and 2023. Even so, the SACU average (0.39 in 2023) remains above the advanced-economy average (0.36) and close to the emerging-market level (0.37). The progress reflects both rising spending and strengthened investment-management frameworks, but the remaining gap points to scope for further gains.

Health gaps are persistent and larger than peers. The middle panel shows health efficiency gaps of 0.36 to 0.49 for SACU countries, little changed since 2000 (when the average gap was 0.42 compared to 0.41 in 2023), and above the peer-group averages. Health outcomes in SACU continue to reflect the lingering effects of the HIV/AIDS epidemic, but also structural features of the health systems—urban-rural coverage gaps, workforce pressures, and supply-chain weaknesses—that limit how efficiently spending translates into outcomes.

Education gaps have been broadly stable. The right panel shows education efficiency gaps of 0.31 to 0.44 across SACU. Consequently, the SACU average gap (0.35 in 2023) is in line with the emerging-market average (0.35) and below the sub-Saharan Africa average (0.37). Most countries experienced limited change in efficiency between 2000 and 2023. Lesotho stands out as the exception, with its gap narrowing notably from 0.44 to 0.33. The gaps in South Africa and Namibia ended modestly above their 2000 levels, reaching 0.41 and 0.37 respectively.

1. R&D lies outside the scope of this paper and is not considered further; the analysis focuses on public investment, education, and health.

Figure 4: Efficiency Gaps in Health, Education, and Infrastructure Spending
(Comparison between 2000 and 2023)



Source: IMF staff estimates based on IMF (2025a).
Note: The chart compares 2000 and 2023 values across three sectors. The thick dark-grey line is the SACU aggregate; thin solid lines show individual countries; dashed/dotted lines show reference averages. The y-axis range is 0.2–0.6; Lesotho's off-chart 2000 value (0.88) is annotated.

Spending-efficiency comparisons reveal notable differences in where SACU countries stand internationally. Figure 5 sets these gaps in global context, plotting the 2023 efficiency gaps for every country in the database across the three sectors, ordered from highest to lowest gap. In infrastructure (top panel), Namibia, Lesotho, and Eswatini sit close to the median of the global distribution, while Botswana and South Africa are below the median. In contrast, health efficiency gaps in SACU are among the widest in the world (middle panel); all five countries sit in the extreme left tail of the global distribution, with Lesotho (2nd largest gap globally), South Africa (5th), Botswana (6th), Eswatini (9th), and Namibia (14th) highlighting severe inefficiencies. In education (bottom panel), the dispersion is similar to infrastructure: South Africa and Namibia sit in the upper part of the global gap distribution, while Botswana, Lesotho, and Eswatini are closer to the global median.

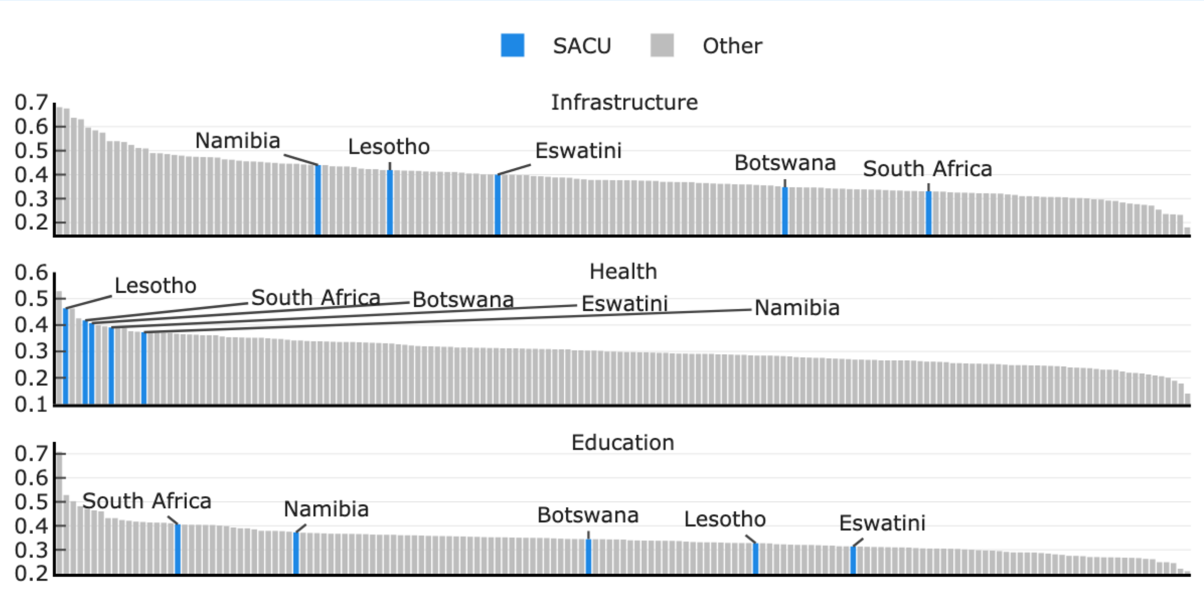
Table 1 summarizes the country-specific efficiency gaps that feed the simulation exercise in Sections 5 and 6. These represent the latest available estimates from the database. For human capital, we report both the education-specific and the health-specific gaps, as well as a blended gap defined as the simple average of the two. The blended gap is used in the model because the Fiscal Monitor DSGE framework treats education and health as a single public human-capital stock (schools and hospitals).

Table 1: SACU Efficiency Gaps, Latest Available Year
(Efficiency gap, 0 to 1 scale; higher values indicate larger gaps. Used as inputs to the DSGE simulations)

Country	Infrastructure (e_{GI}, 2023)	Education (e_{GH}^{edu}, 2024)	Health (e_{GH}^{hlt}, 2023)	Blended HC (e_{GH}^{blend})
Botswana	0.35	0.36	0.41	0.38
Eswatini	0.40	0.32	0.39	0.36
Lesotho	0.42	0.34	0.46	0.40
Namibia	0.44	0.39	0.37	0.38
South Africa	0.33	0.42	0.42	0.42

Source: IMF Fiscal Monitor (2025) efficiency-score database, extracted for the FM2025_SACU_DSGE model.
Note: Blended human-capital gap is the simple average of the education and health gaps.

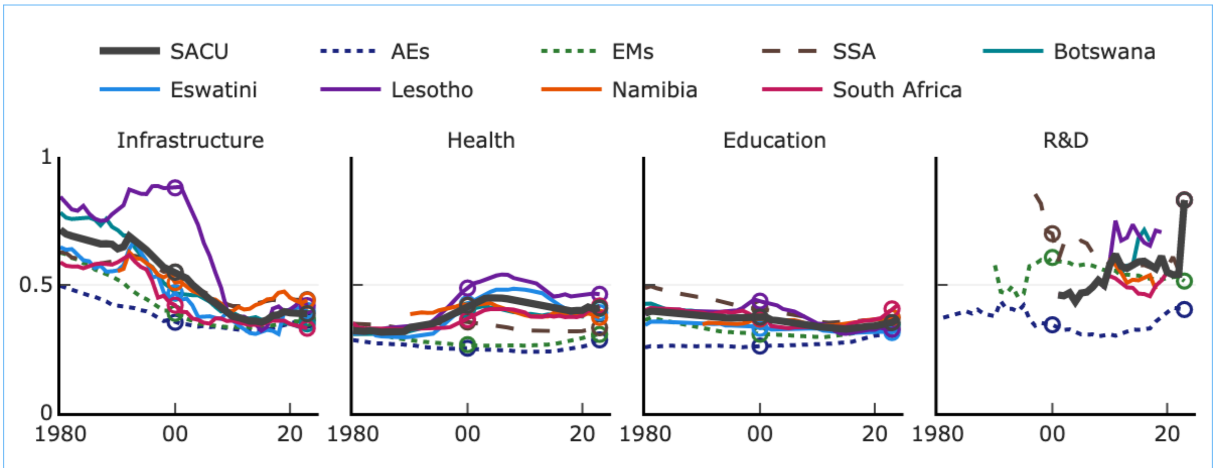
Figure 5: Spending-Efficiency Gaps Across Countries, 2023
(Cross-country distribution of spending-efficiency gaps in three sectors, with SACU countries highlighted)



Source: IMF staff estimates based on IMF (2025a).
Note: Each bar is one country; SACU members highlighted in blue.

A Appendix: Spending-Efficiency Gaps over Time

Figure 6: SACU Spending-Efficiency Gaps over Time
(Efficiency gap, 0–1 scale; higher values indicate larger gaps)



Source: IMF staff estimates based on IMF (2025a).
Note: The thick dark-grey line is the SACU aggregate (unweighted mean of the five members); thin solid lines show the individual countries; dotted lines show reference averages—advanced economies (AEs), emerging markets (EMs), and sub-Saharan Africa (SSA). Hollow circles mark 2000 and 2023.

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